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Federated Adaptive Causal Estimation (FACE) of Target Treatment Effects

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ABSTRACT

Federated learning of causal estimands may greatly improve estimation efficiency by leveraging data from multiple study sites, but robustness to heterogeneity and model misspecifications is vital for ensuring validity. We develop a Federated Adaptive Causal Estimation (FACE) framework to incorporate heterogeneous data from multiple sites to provide treatment effect estimation and inference for a flexibly specified target population of interest. FACE accounts for site-level heterogeneity in the distribution of covariates through density ratio weighting. To safely incorporate source sites and avoid negative transfer, we introduce an adaptive weighting procedure via a penalized regression, which achieves both consistency and optimal efficiency. Our strategy is communication-efficient and privacy-preserving, allowing participating sites to share summary statistics only once with other sites. We conduct both theoretical and numerical evaluations of FACE and apply it to conduct a comparative effectiveness study of BNT162b2 (Pfizer) and mRNA-1273 (Moderna) vaccines on COVID-19 outcomes in U.S. veterans using electronic health records from five VA regional sites. We show that compared to traditional methods, FACE meaningfully increases the precision of treatment effect estimates, with reductions in standard errors ranging from 26% to 67%. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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1. Introduction

Multi-center, federated causal inference is of great interest, particularly when studying novel treatments, rare diseases, or in times of urgent health crises. For example, the COVID-19 pandemic has highlighted the need for novel approaches to efficiently and safely evaluate the effectiveness of novel therapies and vaccines, while leveraging data from multiple healthcare systems to ensure the generalizability of findings. Over the past few years, many research networks and data consortia have been built to facilitate multi-site studies and have been actively contributing to COVID-19 studies, including the Observational Health Data Sciences and Informatics (OHDSI) consortium (Hripcsak et al. 2016) and the Consortium for Clinical Characterization of COVID-19 by Electronic Health Records (EHR) (Brat et al. 2020).

Analyzing data collected from multiple healthcare systems, however, is highly challenging for several reasons. Various sources of heterogeneity exist in terms of (i) differences in the underlying population of each dataset and (ii) policy-level variations of treatment assignment. Since treatment effects may differ across different patient populations, it would be of interest to infer the average treatment effect (ATE) for specific target populations. However, the presence of heterogeneity and potential model misspecification poses great difficulty in ensuring valid estimates for the target average treatment

effect (TATE). Furthermore, patient-level data typically cannot be shared across healthcare centers, which brings additional practical challenges. To overcome these challenges, we propose a Federated Adaptive Causal Estimation (FACE) framework that aims to incorporate heterogeneous data from multiple sites to make inferences about the TATE, while accounting for heterogeneity and data-sharing constraints.

Most existing literature on federated learning has focused on regression and classification models (Chen et al. 2006; Li, Lin, and Li 2013; Chen and Xie 2014; Lee et al. 2017; Lian and Fan 2017; Wang et al. 2019; Duan et al. 2019). Limited federated learning methods currently exist to make causal inferences with multiple heterogeneous studies. Recently, Xiong et al. (2021) proposed federated inverse probability weighted (IPW) estimation of the ATE specifically for an entire study population. Although Xiong et al. (2021) provided multiple methods for point estimation and variance estimation, the choice of the proper method depends on prior knowledge about model homogeneity and specification, which are difficult to verify in practice. No empirical study in Xiong et al. (2021) was provided to test the robustness of the approach to the covariate shift assumption. In addition, their methods cannot be used to estimate the ATE of a target population that differs from the full study population. Vo et al. (2021) proposed a Bayesian approach that models potential outcomes as random functions distributed

by Gaussian processes. Their focus is also on the population ATE rather than any particular target population, and their approach requires specifying parameters and hyperparameters of Gaussian processes and modeling between-site covariate correlations through kernel functions, which can be numerically intensive. Compared to these approaches, our approach estimates the TATE in a particular target population and accounts for the heterogeneity across populations without requiring prior information on the source data distribution or the validity of model specifications. Our approach further safeguards against incorporating source datasets that may introduce bias to the TATE estimate, known as negative transfer (Pan and Yang 2009; Weiss, Khoshgoftaar, and Wang 2016).

Another related strand of literature concerns the generalizability and transportability of randomized clinical trials to EHR studies. For example, Stuart et al. (2011) and Stuart, Bradshaw, and Leaf (2015) assessed the generalizability of results from randomized trials to target populations of interest. Dahabreh et al. (2020), Josey et al. (2022), and Lee et al. (2023) all focused on extending inferences about treatments from a randomized trial to a new target population by using different weighting schemes. For a comprehensive review of statistical methods for generalizability and transportability, see Degtiar and Rose (2023). However, to date, no literature in generalizability and transportability has sought to leverage observational data from a potentially large number of source sites in a data-adaptive manner to obtain unbiased, efficient, and robust estimation of target treatment effects.

In the context of statistical inference, negative transfer occurs when incorporating a source dataset increases the bias or asymptotic variance of the estimator compared to not including it. Negative transfers can arise from several reasons, including differences in the distribution of covariates (covariate shift), distribution shifts that result in differences in the conditional average treatment effects (CATE), and differences in the distribution of unmeasured confounders. FACE adjusts for covariate shift by estimating density ratio weights that satisfy a balancing condition that aligns the source site covariate distribution with that of the target site. FACE features an adaptive integration step where site-specific weights are learned to control the contribution of source sites to minimize the variance of the federated estimator while accounting for potential bias. When there are significant differences in CATEs between the target and source sites, the weights are adaptively shrunk toward zero, reducing the influence of non-informative source sites. In doing so, FACE can achieve optimal efficiency while maintaining consistency, and it is robust to the distribution of data and potential model misspecifications in the source sites. Moreover, FACE is a communication-efficient federated algorithm that allows each participating site to keep its data stored locally and only share summary statistics once with other sites. Negative transfers can also be caused by unmeasured confounders that differ between target and source sites that must be controlled for to achieve ignorability. FACE does not directly handle this source of negative transfer, and in such scenarios, alternative approaches like sensitivity analyses or methods designed to handle unobserved distribution shifts may be useful (Andrews and Oster 2017; Nguyen et al. 2017; Nie, Imbens, and Wager 2021).

The major contributions of FACE can be summarized as follows. First, FACE allows for flexibility in the specification of the target population. For example, the target population in a research network can be defined as the underlying population of a given healthcare center, or multiple healthcare centers that share certain properties (e.g., geographic location), or the overall population combining all sites. This flexibility provides stakeholders and policymakers at different levels with information on their respective target populations. Second, using a semiparametric density ratio weighting approach, FACE allows the distribution of covariates to be heterogeneous across sites. Third, FACE protects against negative transfer through an adaptive integration strategy that anchors on the target data and computes data-adaptive weights for source sites.

The remainder of the article is organized as follows. In Section 2, we introduce the problem setting, notation, and assumptions required for identification of the TATE. In Section 3, we describe the proposed FACE framework for estimating the TATE. We introduce the in-site estimators based on the target population and source populations separately in Sections 3.1 and 3.2, present the adaptive and distributed integration in Section 3.3, a cross-validation approach to select tuning parameters in Section 3.4, and describe FACE-X, a variant of FACE that incorporates nonparametric machine learning methods to estimate nuisance functions in Section 3.5. In Section 4, we provide the theoretical guarantees of FACE, including double robustness, asymptotic normality, and relative efficiency. In Section 5, we conduct extensive simulations for various data generating mechanisms and show robustness to misspecification of different models. In Section 6, we apply FACE to conduct a comparative effectiveness study of COVID-19 vaccines using the EHRs from five geographic regions of the Department of Veterans Affairs (VA). We conclude in Section 7 with key takeaways and directions for future research.

2. Setting and Notation

For the i th observation, we denote the outcome as $Y_i \in \mathbb{R}$, the p -dimensional baseline covariate vector as $\mathbf{X}_i = (X_{i1}, \dots, X_{ip})^\top \in \mathcal{X} \subset \mathbb{R}^p$, and the indicator for binary treatment as $A_i \in \{0, 1\}$. There are $J \geq 1$ target sites and another $K \geq 0$ source sites. Let $\mathcal{T} \subseteq [J + K]$ indicate sites that are in the target population and $\mathcal{S} \subseteq [J + K]$ indicate sites that are in the source population, where $[K] = \{1, \dots, K\}$ for any integer K . Under the federated learning setting, a total of N observations are stored at $J + K$ study sites, where the k th site has sample size n_k , and $N = \sum_{k=1}^{J+K} n_k$. Let R_i be a site indicator such that $R_i = k$ indicates the i th patient in the k th site. Indexing the site by a single integer R_i , we assume that each observation may only belong to one site. We summarize the observed data at each site k as $\mathcal{D}_k = \{(Y_i, \mathbf{X}_i^\top, A_i, R_i)^\top, R_i = k\}$, and consider a federated data setting where each site has access to its own patient-level data but can share only summary statistics with other sites. We denote the index set for each site as $\mathcal{I}_k = \{i : R_i = k\}$. The data included in the target sites are denoted by $\mathcal{D}_{\mathcal{T}}$. For simplicity of notation, we use $(Y, \mathbf{X}, A, R)$ without subscripts to state general assumptions and conclusions.

Under the potential outcomes framework (Neyman 1923; Rubin 1974), we denote $Y^{(a)}$ as the potential outcome of patients under treatment $A = a$, $a = 0, 1$. Our goal is to estimate the TATE for a specified target population $\mathcal{T}$,

$$\Delta_{\mathcal{T}} = \mu_{\mathcal{T}}^{(1)} - \mu_{\mathcal{T}}^{(0)}, \quad \mu_{\mathcal{T}}^{(a)} = \mathbb{E}(Y^{(a)} \mid R \in \mathcal{T}), \quad (1)$$

where the expectation is taken over the distribution in the target population. The target population can be specified at multiple levels (e.g., single site, multiple sites, all sites) corresponding to different targets of real-world interest. This distinction between target and source sites also distinguishes our setting from that of Xiong et al. (2021), in which the target population always contains all participating sites.

To identify the TATE, we make the following standard assumptions (Imbens and Rubin 2015; Hernán and Robins 2020) throughout the article:

Assumption 1. For a positive constant $\varepsilon > 0$, $a \in \{0, 1\}$, and $\mathbf{x} \in \mathcal{X}$,

- (a) Consistency: $Y = Y^{(A)}$.
- (b) Overlapping of treatment arms: $\mathbb{P}(A = a \mid \mathbf{X} = \mathbf{x}, R = k) \in (\varepsilon, 1 - \varepsilon)$, $k \in [J + K]$.
- (c) Overlapping of site populations: $\mathbb{P}(R = k \mid \mathbf{X} = \mathbf{x}) > \varepsilon$, $k \in [J + K]$.
- (d) Ignorability: $(Y^{(1)}, Y^{(0)}) \perp\!\!\!\perp (A, R) \mid \mathbf{X}$ for $R \in \{\mathcal{T}, \mathcal{S}^*\}$ for some $\mathcal{S}^* \subseteq \mathcal{S}$.

Remark 1. Assumption 1(d) implies that the underlying true treatment response pattern is shared across target sites and an unspecified subset of source sites $\mathcal{S}^* \subseteq \mathcal{S}$ so that the treatment effect estimates from $\mathcal{T}$ and $\mathcal{S}^*$ can be safely combined to estimate the TATE. Our adaptive selection and aggregation step in FACE, as detailed in Section 3.3, is designed to incorporate these informative source sites $\mathcal{S}^*$ for precision gain while preventing negative transfer from non-informative source sites $\mathcal{S} \setminus \mathcal{S}^*$. Assumption 1(d) assumes that controlling for observed confounders is sufficient and is similar to assumption C1 made in Dehejia, Pop-Eleches, and Samii (2021). Assumption 1(d) may be violated, for example, when the target and source populations differ along unobserved features. Nie, Imbens, and Wager (2021) considered such a setting by assuming that the distribution of potential outcomes across target and source populations are the same conditioning on observed confounders $\mathbf{X}$ and unmeasured effect modifiers $\mathbf{U}$ and derive bounds for the TATE by assuming a sensitivity model that directly implies a bound on the unobserved distribution shift ratio. Since violations of the transportability assumption are in general untestable, many works have also proposed sensitivity analysis for how much violation of the assumption can result in transportability bias (Andrews and Oster 2017; Nguyen et al. 2017).

We denote the specified models for the site-specific propensity score (PS) and outcome regression (OR) as:

$$\text{PS:} \quad \mathbb{P}(A = a \mid R = k, \mathbf{X}) = \pi_k(a, \mathbf{X}; \boldsymbol{\alpha}_k), \quad (2)$$

$$\text{OR:} \quad \mathbb{E}(Y \mid R = k, A = a, \mathbf{X}) = m(a, \mathbf{X}; \boldsymbol{\beta}_{a,k}). \quad (3)$$

For the target sites, we require $E(Y^{(a)} \mid R = k, \mathbf{X})$ to be shared but do not require $\boldsymbol{\alpha}_k$ to be the same across $\mathcal{T}$. Under possible

model misspecifications, we allow either (i) the outcome models in (3) to be correctly specified with $\boldsymbol{\beta}_{a,k} = \boldsymbol{\beta}_a$, or (ii) the PS models in (2) to be correctly specified, for $k \in \mathcal{T}$.

Since the distribution of the covariates $\mathbf{X}$ can be heterogeneous across sites, we characterize the difference in covariate distributions between a target site $k_t \in \mathcal{T}$ and a source site $k_s \in \mathcal{S}$ through a density ratio

$$\omega_{k_t, k_s}(\mathbf{x}) = \frac{f(\mathbf{X} \mid R = k_t)}{f(\mathbf{X} \mid R = k_s)} = \frac{\mathbb{P}(R = k_t \mid \mathbf{X} = \mathbf{x})\mathbb{P}(R = k_s)}{\mathbb{P}(R = k_s \mid \mathbf{X} = \mathbf{x})\mathbb{P}(R = k_t)}.$$

We choose flexible semiparametric models for the density ratio

$$\omega_{k_t, k_s}(\mathbf{X}; \boldsymbol{\gamma}_{k_t, k_s}) = \exp\{\boldsymbol{\gamma}_{k_t, k_s}^\top \boldsymbol{\psi}(\mathbf{X})\}, \quad (4)$$

where $\boldsymbol{\psi} : \mathbb{R}^p \mapsto \mathbb{R}^q$ is a vector-valued basis function with an intercept term. One may specify a range of basis functions to capture potential non-linearity in the density ratio model to improve the robustness of the estimation for $\omega_{k_t, k_s}(\mathbf{x})$.

Remark 2. The exponential tilt density ratio model (4) is widely used to account for heterogeneity between two distributions (Qin 1998; Qin and Liang 2011; Duan et al. 2020b). By including higher-order terms of $\mathbf{x}$ in $\boldsymbol{\psi}(\mathbf{x})$, higher-order differences such as variance and skewness can be captured. We propose in Section 3 a communication-efficient approach to estimate $\boldsymbol{\gamma}_{k_t, k_s}$ in covariate distributions between a target site and source site without sharing individual-level data. In the simulation study and real-data example, we have selected the exponential tilt model with $\boldsymbol{\psi}(\mathbf{x}) = \mathbf{x}$, which recovers the whole class of natural exponential family distributions, including the normal distribution with mean shift, Bernoulli distribution for binary covariates, etc. More flexible choices for $\boldsymbol{\psi}(\cdot)$ could help calibrate higher-order moments of covariates. However, if the number of covariates p is high, one must consider the tradeoff between the amount of information that is shared and the feasibility of balancing covariate distributions.

3. FACE Method

In this section, we detail the FACE method. We start with an overview of its main workflow, where a schematic illustration can be found in Figure S1 of the supplementary materials. In Step 1, each target site calculates summary statistics of its covariate distribution, $\bar{\boldsymbol{\psi}}_k = n_k^{-1} \sum_{i \in \mathcal{I}_k} \boldsymbol{\psi}(\mathbf{X}_i)$ for $k \in \mathcal{T}$, a key quantity for estimating the density ratio model to balance covariate distributions, and broadcasts them to all source sites, along with its OR parameters $\{\hat{\boldsymbol{\beta}}_{a,k}, a = 0, 1\}$. Each target site also constructs a doubly robust estimator for its site-specific ATE, obtains additional summary statistics needed for the adaptive aggregation, and shares them with the leading analysis center (AC) (see Section 3.1). In Step 2, each source site uses the summary statistics of the target site ($\bar{\boldsymbol{\psi}}_k$ from $k \in \mathcal{T}$) to fit its density ratio model and construct an augmentation term $\hat{\delta}_{\mathcal{T}, k}$ for $k \in \mathcal{S}$ for the TATE. Each source site shares the augmentation term, together with additional summary statistics needed for the aggregation, to the AC (see Section 3.2). In Step 3, the AC performs the aggregation with estimators and parameters from Steps 1 and 2 to obtain the final FACE estimator, $\hat{\Delta}_{\mathcal{T}, \text{FACE}}$ (see Section 3.3). Overall, each site is only required to share information one time with other sites.

We detail each step of FACE in Sections 3.1–3.3 with generic models. Each site will need to fit both the OR model and the PS model using its own local data. Standard regression models such as logistic regression and generalized linear models can be used. Nonlinear basis functions can be included to incorporate nonlinear effects. For $k \in [J + K]$, we denote the estimated PS as $\pi_k(a, \mathbf{X}; \hat{\alpha}_k)$ and the predicted outcome for treatment a as $m(a, \mathbf{X}; \hat{\beta}_{a,k})$, where $\hat{\alpha}_k$ and $\hat{\beta}_{a,k}$ can be achieved via classical estimation methods such as maximum likelihood estimation or estimating equations. An example with logistic regression models is given in Section C of the supplementary materials.

3.1. Step 1: Estimation Using Target Data

The initial doubly robust TATE estimator is obtained from the site-specific ATE of the target sites. Within target sites $k \in \mathcal{T}$, we compute the doubly robust TATE (Bang and Robins 2005), $\hat{\Delta}_{\mathcal{T},k} = \hat{M}_k + \hat{\delta}_{\mathcal{T},k}$, where

$$\hat{M}_k = n_k^{-1} \sum_{i \in \mathcal{I}_k} \{m(1, \mathbf{X}_i; \hat{\beta}_{1,k}) - m(0, \mathbf{X}_i; \hat{\beta}_{0,k})\} \text{ for } k \in \mathcal{T}$$

is the OR model based estimate of the TATE, and

$$\hat{\delta}_{\mathcal{T},k} = n_k^{-1} \sum_{i \in \mathcal{I}_k} \frac{(-1)^{1-A_i}}{\pi_k(A_i, \mathbf{X}_i; \hat{\alpha}_k)} \{Y_i - m(A_i, \mathbf{X}_i; \hat{\beta}_{A_i,k})\} \text{ for } k \in \mathcal{T} \quad (5)$$

is the augmentation term that guards against misspecification of the OR model. In addition, we calculate summary statistics for the $k \in \mathcal{T}$ target site covariate distribution, $\bar{\psi}_k = n_k^{-1} \sum_{i \in \mathcal{I}_k} \psi(\mathbf{X}_i)$. The AC can construct the initial TATE estimate,

$$\hat{\Delta}_{\mathcal{T},\mathcal{T}} = N_{\mathcal{T}}^{-1} \sum_{k \in \mathcal{T}} n_k \hat{\Delta}_{\mathcal{T},k}$$

with summary data from target sites, $\{\hat{\Delta}_{\mathcal{T},k}, n_k : k \in \mathcal{T}\}$. The consistency of $\hat{\Delta}_{\mathcal{T},\mathcal{T}}$ is ensured when either the PS or OR is consistently estimated for each $k \in \mathcal{T}$.

Remark 3. Here, we estimate β_a in each target site $k \in \mathcal{T}$ as $\hat{\beta}_{a,k}$. Alternatively, one could estimate β_a jointly at the cost of one additional round of communication between target sites. A jointly estimated β_a could benefit from efficiency gain under certain model specification conditions. Previous literature have developed distributed methods for aggregating estimates of β_a (Chen et al. 2006; Huang and Huo 2019; Duan et al. 2020a). In practice, one should balance the advantage of potential efficiency gain with the cost of additional cross-site communication.

To facilitate optimal aggregation, we also share the estimators for the variance-covariance of scaled estimators, $\sqrt{n_k}(\hat{M}_k, \hat{\delta}_{\mathcal{T},k}, \bar{\psi}_k, \hat{\beta}_{1,k}, \hat{\beta}_{0,k})$, which we denote as $\hat{\Sigma}_k$ for the target sites $k \in \mathcal{T}$. Variance estimation $\hat{\Sigma}_k$ for $k \in \mathcal{T}$ can be conducted through classical influence functions or bootstrapping within site. The exact role of the matrix in the aggregation will be unveiled after introducing the optimal combination weights in (11), which is the centerpiece of the adaptive aggregation step.

3.2. Step 2: Estimation Using Source Data

To safely use source data to assist in estimating $\Delta_{\mathcal{T}}$, we further account for the covariate shifts between the source sites and the target sites by tilting the source sites to the target population through the density ratios $\omega_{k_t,k_s}(\mathbf{X}; \gamma_{k_t,k_s})$. If individual-level data could be shared, estimating $\hat{\gamma}_{k_t,k_s}$ could be achieved by constructing a pseudo-likelihood function as in Qin (1998). However, such an estimator cannot be directly obtained in a federated data setting. Instead, we propose a simple estimating equation approach that can be calculated in each source site $k_s \in \mathcal{S}$ using its data, along with summary statistics $\bar{\psi}_{k_t}$ obtained from the target sites $k_t \in \mathcal{T}$. Specifically, we estimate γ_{k_t,k_s} as

$$\hat{\gamma}_{k_t,k_s} : \text{solution to } n_{k_s}^{-1} \sum_{i \in \mathcal{I}_{k_s}} \omega_{k_t,k_s}(\psi(\mathbf{X}_i); \gamma_{k_t,k_s}) \psi(\mathbf{X}_i) = \bar{\psi}_{k_t}. \quad (6)$$

Remark 4. To ensure the existence of a solution, consider the density ratio model with linear predictor $\omega_{k_t,k_s}(\psi(\mathbf{X}_i); \gamma_{k_t,k_s}) = \omega_{k_t,k_s}(\gamma_{k_t,k_s}^\top \psi(\mathbf{X}_i))$ and monotonic link function with a positive derivative, $\omega'_{k_t,k_s}(x) > 0$. Define the convex antiderivative of the link function as $\Omega_{k_t,k_s}(x)$, such that $\Omega'_{k_t,k_s}(x) = \omega_{k_t,k_s}(x)$. This allows us to reformulate the estimating equation as a convex optimization problem:

$$\hat{\gamma}_{k_t,k_s} = \arg \min_{\gamma} \left\{ n_{k_s}^{-1} \sum_{i \in \mathcal{I}_{k_s}} \Omega_{k_t,k_s}(\gamma^\top \psi(\mathbf{X}_i)) - \gamma^\top \bar{\psi}_{k_t} \right\}. \quad (7)$$

The convexity of Ω_{k_t,k_s} ensures that the objective function is convex, and under mild regularity conditions, a unique global minimum exists. Specifically, the solution exists whenever the Hessian matrix is positive definite:

$$\inf_{\|\mathbf{v}\|_2=1} n_{k_s}^{-1} \sum_{i \in \mathcal{I}_{k_s}} \omega'_{k_t,k_s}(\gamma^\top \psi(\mathbf{X}_i)) \left\{ \mathbf{v}^\top \psi(\mathbf{X}_i) \right\}^2 > 0.$$

Standard regularity conditions on the lower bound for the eigenvalues of the asymptotic Hessian,

$$\inf_{\|\mathbf{v}\|_2=1} \mathbf{v}^\top \mathbb{E} \left[\omega'_{k_t,k_s}(\gamma^\top \psi(\mathbf{X}_i)) \psi(\mathbf{X}_i) \psi(\mathbf{X}_i)^\top \mid R = k_s \right] \mathbf{v} > 0,$$

guarantees the existence of a solution when n_{k_s} is sufficiently large. In finite samples, an exact solution to the balancing equation may not exist. To address this, we can introduce a penalized optimization problem:

$$\hat{\gamma}_{k_t,k_s} = \arg \min_{\gamma} \left\{ n_{k_s}^{-1} \sum_{i \in \mathcal{I}_{k_s}} \Omega_{k_t,k_s}(\gamma^\top \psi(\mathbf{X}_i)) - \gamma^\top \bar{\psi}_{k_t} + \lambda_{\gamma} \sum_{j=1}^p \rho_j |\gamma_j| \right\}, \quad (8)$$

where $\lambda_{\gamma} > 0$ is a tuning parameter controlling the regularization strength and ρ_j are scaling factors for each component γ_j . The optimality conditions lead to approximate balancing

constraints resembling those in approximate balancing weights methods (Wang and Zubizarreta 2020),

$$\left| n_{k_s}^{-1} \sum_{i \in \mathcal{I}_{k_s}} \omega_{k_t, k_s}(\boldsymbol{\psi}(\mathbf{X}_i); \boldsymbol{\gamma}_{k_t, k_s}) \psi_j(\mathbf{X}_i) - \bar{\psi}_{k_t, j} \right| \leq \rho_j \lambda_{\boldsymbol{\gamma}}.$$

The factors ρ_j can be chosen to harmonize the scale of $\psi_j(\mathbf{X}_i)$ or to reflect their importance for the nuisance models.

Remark 5. Our approach is related to recent work that adjusts for observed differences in covariate distributions between a target population and the population that actually receives treatments (Hirshberg, Maleki, and Zubizarreta 2019; Tan 2020). Hirshberg, Maleki, and Zubizarreta (2019) construct minimax linear weights that achieve approximate sample balance as in (6) uniformly over an absolutely convex class $\mathcal{M}$. They show that when $\mathcal{M}$ is selected appropriately, the solution to (6) converges in empirical mean square to the functional's Riesz representer, that is, the unique square-integrable function that satisfies the corresponding population balance condition for all square-integrable functions (Hirshberg and Wager 2021). Relatedly, Tan (2020) propose regularized calibrated estimators in the high-dimensional setting under minimal sparsity assumptions.

For each source site, we construct a site augmentation term similar to the augmentation term in (5) for the target sites but with an additional density ratio weight

$$\hat{\delta}_{\mathcal{T}, k_s} = N_{\mathcal{T}}^{-1} \sum_{k_t \in \mathcal{T}} \frac{n_{k_t}}{n_{k_s}} \sum_{i \in \mathcal{I}_{k_s}} \omega_{k_t, k_s}(\mathbf{X}_i; \hat{\boldsymbol{\gamma}}_{k_t, k_s}) \frac{(-1)^{1-A_i}}{\pi_{k_s}(A_i, \mathbf{X}_i; \hat{\boldsymbol{\alpha}}_{k_s})} \{Y_i - m(A_i, \mathbf{X}_i; \hat{\boldsymbol{\beta}}_{A_i, k_t})\} \text{ for } k_s \in \mathcal{S}.$$

We use the OR estimates from target sites $\hat{\boldsymbol{\beta}}_{A_i, k_t}$ to ensure robustness when the OR is misspecified. See Remark 6 for details.

Then, the site-specific augmentation terms $\hat{\delta}_{\mathcal{T}, k_s}$ are shared back to the AC, together with (i) $\hat{\sigma}_{k_s}^2$, an estimate for the scaled conditional variance $n_{k_s} \text{var}(\hat{\delta}_{\mathcal{T}, k_s} | \mathcal{D}_{\mathcal{T}})$, and (ii) $\hat{\mathbf{d}}_{k_t, k_s}$, an estimate for the partial derivatives of $\hat{\delta}_{\mathcal{T}, k_s}$ with respect to $\hat{\boldsymbol{\psi}}_{k_t}$, $\hat{\boldsymbol{\beta}}_{1, k_t}$, and $\hat{\boldsymbol{\beta}}_{0, k_t}$. The role of $\hat{\mathbf{d}}_{k_t, k_s}$ in the aggregation will be explained in (11). Both $\hat{\sigma}_{k_s}^2$ and $\hat{\mathbf{d}}_{k_t, k_s}$ can be constructed from classical influence functions. Alternatively, $\hat{\sigma}_{k_s}^2$ can be estimated by bootstrapping within site and $\hat{\mathbf{d}}_{k_t, k_s}$ can be estimated by numerical derivatives.

Remark 6. Combining the source site augmentation term $\hat{\delta}_{\mathcal{T}, k_s}$ with the initial TATE OR estimator from the target sites $\hat{M}_{\mathcal{T}} = N_{\mathcal{T}}^{-1} \sum_{k_t \in \mathcal{T}} n_{k_t} \hat{M}_{k_t}$, we obtain the $k_s \in \mathcal{S}$ source site estimators $\hat{\Delta}_{\mathcal{T}, k_s} = \hat{M}_{\mathcal{T}} + \hat{\delta}_{\mathcal{T}, k_s}$ as

$$\hat{\Delta}_{\mathcal{T}, k_s} = N_{\mathcal{T}}^{-1} \sum_{k_t \in \mathcal{T}} n_{k_t} \left(n_{k_t}^{-1} \sum_{i \in \mathcal{I}_{k_t}} \{m(1, \mathbf{X}_i; \hat{\boldsymbol{\beta}}_{1, k_t}) - m(0, \mathbf{X}_i; \hat{\boldsymbol{\beta}}_{0, k_t})\} \right)$$

$$+ n_{k_s}^{-1} \sum_{i \in \mathcal{I}_{k_s}} \omega_{k_t, k_s}(\mathbf{X}_i; \hat{\boldsymbol{\gamma}}_{k_t, k_s}) \frac{(-1)^{1-A_i}}{\pi_{k_s}(A_i, \mathbf{X}_i; \hat{\boldsymbol{\alpha}}_{k_s})} \{Y_i - m(A_i, \mathbf{X}_i; \hat{\boldsymbol{\beta}}_{A_i, k_t})\}.$$

When the underlying OR model in the $k_s \in \mathcal{S}$ source site is the same as in the target population, the estimator $\hat{\Delta}_{\mathcal{T}, k_s}$ is doubly robust in the following sense: either (i) the OR model is consistent for all $k \in \{\mathcal{T}, k_s\}$, or (ii) the PS and density ratio models are consistent for the source site. Shifts in covariate distributions may induce heterogeneity in OR estimates across sites under misspecified OR models, even if the conditional distribution $Y | A, \mathbf{X}$ is shared. To achieve robustness against misspecified OR models, it is important to use the same $\hat{\boldsymbol{\beta}}_{a, k_t}$ for $\hat{M}_{\mathcal{T}}$ and $\hat{\delta}_{\mathcal{T}, k_s}$ so that we may rely on the correct PS and density ratio models for consistency according to the alternative representation

$$N_{\mathcal{T}}^{-1} \sum_{k_t \in \mathcal{T}} n_{k_t} \left\{ n_{k_s}^{-1} \sum_{i \in \mathcal{I}_{k_s}} \omega_{k_t, k_s}(\mathbf{X}_i; \hat{\boldsymbol{\gamma}}_{k_t, k_s}) \frac{(-1)^{1-A_i}}{\pi_{k_s}(A_i, \mathbf{X}_i; \hat{\boldsymbol{\alpha}}_{k_s})} Y_i + n_{k_t}^{-1} \sum_{i \in \mathcal{I}_{k_t}} m(1, \mathbf{X}_i; \hat{\boldsymbol{\beta}}_{1, k_t}) - n_{k_s}^{-1} \sum_{i \in \mathcal{I}_{k_s}} \omega_{k_t, k_s}(\mathbf{X}_i; \hat{\boldsymbol{\gamma}}_{k_t, k_s}) \frac{A_i}{\pi_{k_s}(1, \mathbf{X}_i; \hat{\boldsymbol{\alpha}}_{k_s})} m(1, \mathbf{X}_i; \hat{\boldsymbol{\beta}}_{1, k_t}) - n_{k_t}^{-1} \sum_{i \in \mathcal{I}_{k_t}} m(0, \mathbf{X}_i; \hat{\boldsymbol{\beta}}_{0, k_t}) + n_{k_s}^{-1} \sum_{i \in \mathcal{I}_{k_s}} \omega_{k_t, k_s}(\mathbf{X}_i; \hat{\boldsymbol{\gamma}}_{k_t, k_s}) \frac{1 - A_i}{\pi_{k_s}(0, \mathbf{X}_i; \hat{\boldsymbol{\alpha}}_{k_s})} m(0, \mathbf{X}_i; \hat{\boldsymbol{\beta}}_{0, k_t}) \right\}.$$

To protect against negative transfer from source sites with biased TATE estimators, we combine information from each source site with the target sites through our adaptive aggregation step in Section 3.3.

3.3. Step 3: Adaptive Aggregation

In the final step, we obtain our FACE estimator by adaptively aggregating the initial TATE estimator $\hat{\Delta}_{\mathcal{T}, \mathcal{T}}$ and the source site estimators $\hat{\Delta}_{\mathcal{T}, k_s}$. Denote $\delta_{\mathcal{T}, \mathcal{T}} = N_{\mathcal{T}}^{-1} \sum_{k \in \mathcal{T}} n_k \delta_{\mathcal{T}, k}$. The AC can estimate $\Delta_{\mathcal{T}}$ by taking a linear combination of the initial TATE estimator $\hat{\Delta}_{\mathcal{T}, \mathcal{T}}$ and the source site estimators $\hat{\Delta}_{\mathcal{T}, k_s}$, where the weights are estimated to make an optimal bias-variance tradeoff. The proposed FACE estimator can be viewed as an “anchor and augmentation” estimator, which weights the source site estimators $\hat{\Delta}_{\mathcal{T}, k_s}$ by η_{k_s} , $k_s \in \mathcal{S}$ and the target estimator $\hat{\Delta}_{\mathcal{T}, \mathcal{T}}$ by $(1 - \sum_{k_s \in \mathcal{S}} \eta_{k_s})$. FACE is given by

$$\begin{aligned} \hat{\Delta}_{\mathcal{T}, \text{FACE}} &= \hat{\Delta}_{\mathcal{T}, \mathcal{T}} + \sum_{k_s \in \mathcal{S}} \eta_{k_s} \{\hat{\Delta}_{\mathcal{T}, k_s} - \hat{\Delta}_{\mathcal{T}, \mathcal{T}}\} \\ &= \hat{\Delta}_{\mathcal{T}, \mathcal{T}} + \sum_{k_s \in \mathcal{S}} \eta_{k_s} \{\hat{\delta}_{\mathcal{T}, k_s} - \hat{\delta}_{\mathcal{T}, \mathcal{T}}\}, \end{aligned} \quad (9)$$

which anchors on the initial TATE estimator $\hat{\Delta}_{\mathcal{T}, \mathcal{T}}$ and is augmented with source site estimators $\hat{\Delta}_{\mathcal{T}, k_s}$, with the weights $\{\eta_{k_s}, k_s \in \mathcal{S}\}$ to be estimated in a data-adaptive fashion to filter out potentially biased source site estimators. The second

expression of $\widehat{\Delta}_{\mathcal{T},\text{FACE}}$ in (9) shows how the parameters from Steps 1 and 2 are used to construct the FACE estimator.

Moreover, the aggregation of the remaining unbiased source site augmentation terms should also minimize the estimation variance. Under the federated learning setting, the key to evaluating the variance of (9) is to decompose it into contributions from separate sites so that they can be estimated within each site. For any subset of $\mathcal{S}$, $\mathcal{S}' \subseteq \mathcal{S}$, we consider the following decomposition

$$\begin{aligned} & \text{var} \left\{ \widehat{\Delta}_{\mathcal{T},\mathcal{T}} + \sum_{k_s \in \mathcal{S}'} \eta_{k_s} (\widehat{\Delta}_{\mathcal{T},k_s} - \widehat{\Delta}_{\mathcal{T},\mathcal{T}}) \right\} \\ & \approx \sum_{k_s \in \mathcal{S}'} \eta_{k_s}^2 \text{var}(\widehat{\delta}_{\mathcal{T},k_s} \mid \mathcal{D}_{\mathcal{T}}) \\ & \quad + \sum_{k_t \in \mathcal{T}} \text{var} \left\{ \left(\frac{n_{k_t}}{N_{\mathcal{T}}}, \frac{n_{k_t} - n_{k_t} \sum_{k_s \in \mathcal{S}'} \eta_{k_s}}{N_{\mathcal{T}}}, \sum_{k_s \in \mathcal{S}'} \eta_{k_s} \mathbf{d}_{k_t,k_s}^{\top} \right) \right. \\ & \quad \left. \left(\widehat{M}_{\mathcal{T}}, \widehat{\delta}_{k_t}, \left(\widehat{\psi}_{k_t}^{\top}, \widehat{\beta}_{1,k_t}^{\top}, \widehat{\beta}_{0,k_t}^{\top} \right) \right)^{\top} \right\}, \end{aligned} \quad (10)$$

where $\mathbf{d}_{k_t,k_s}$ is the limit for $\widehat{\mathbf{d}}_{k_t,k_s}$, which is the partial derivative of $\widehat{\delta}_{\mathcal{T},k_s}$ with respect to the broadcast estimators $\widehat{\psi}_{k_t}$, $\widehat{\beta}_{1,k_t}$ and $\widehat{\beta}_{0,k_t}$. The source site augmentation terms $\widehat{\delta}_{\mathcal{T},k_s}$ involve both the source site $k_s \in \mathcal{S}$ data and estimated parameters $\{\widehat{\psi}_{k_t}, \widehat{\beta}_{1,k_t}, \widehat{\beta}_{0,k_t}\}$ from target sites $k_t \in \mathcal{T}$. We characterize the uncertainty contributions from these two independent sources by $\widehat{\delta}_{\mathcal{T},k_s} \mid \mathcal{D}_{\mathcal{T}}$ and $(\widehat{\psi}_{k_t}^{\top}, \widehat{\beta}_{1,k_t}^{\top}, \widehat{\beta}_{0,k_t}^{\top}) \mathbf{d}_{k_t,k_s}$, respectively. We decouple the dependence of the source site augmentation terms $\widehat{\delta}_{\mathcal{T},k_s}$ on the target sites by subtracting the first order approximation of the dependence $(\widehat{\psi}_{k_t}^{\top}, \widehat{\beta}_{1,k_t}^{\top}, \widehat{\beta}_{0,k_t}^{\top}) \mathbf{d}_{k_t,k_s}$. The resulting $\widehat{\delta}_{\mathcal{T},k_s} - \mathbf{d}_{k_t,k_s}^{\top} \widehat{\psi}_{k_t}$ is asymptotically independent of the target sites.

Since including information from non-informative source sites $\mathcal{S} \setminus \mathcal{S}^*$ may lead to biases, we adopt an adaptive combination strategy similar to the one given in Cheng and Cai (2021) for combining data from a randomized trial and an observational study. Here, we overcome the additional challenge of data sharing constraints, and we propose the following adaptive L_1 penalized optimal aggregation

$$\begin{aligned} \widehat{\eta} = \arg \min_{\eta \in \mathbb{R}^K} N & \underbrace{\left[\sum_{k_s \in \mathcal{S}} \eta_{k_s}^2 \frac{\widehat{\sigma}_{k_s}^2}{n_{k_s}} + \sum_{k_t \in \mathcal{T}} \widehat{\mathbf{h}}_{k_t}(\eta)^{\top} \frac{\widehat{\Sigma}_{k_t}}{n_{k_t}} \widehat{\mathbf{h}}_{k_t}(\eta) \right]}_{\widehat{L}(\eta)} \\ & + \lambda \sum_{k_s \in \mathcal{S}} |\eta_{k_s}| (\widehat{\delta}_{\mathcal{T},k_s} - \widehat{\delta}_{\mathcal{T},\mathcal{T}})^2, \end{aligned} \quad (11)$$

where

$$\widehat{\mathbf{h}}_{k_t}(\eta) = \left(\frac{n_{k_t}}{N_{\mathcal{T}}}, \frac{n_{k_t} - n_{k_t} \sum_{k_s \in \mathcal{S}^*} \eta_{k_s}}{N_{\mathcal{T}}}, \sum_{k_s \in \mathcal{S}} \eta_{k_s} \widehat{\mathbf{d}}_{k_t,k_s}^{\top} \right)^{\top},$$

with $\widehat{\Sigma}_{k_t}$ estimated from Step 1 and $\widehat{\sigma}_{k_s}^2$ and $\widehat{\mathbf{d}}_{k_t,k_s}$ estimated from Step 2. The multiplicative N factor is required to stabilize the loss. Choosing $\lambda \asymp N^{\nu}$ with $\nu \in (0, 1/2)$, we achieve the following oracle property for selection and aggregation: (i)

biased source site augmentation terms have zero weights with high probability; (ii) regularization on the weights for unbiased source site augmentation terms is asymptotically negligible ($\ll N^{-1/2}$). Analogous to the phenomenon in meta-analysis, the estimation uncertainty of $\widehat{\eta}$ has no asymptotic effect on the aggregated estimator.

Using the variance estimator (stabilized by “ N ” factor like-wise)

$$\widehat{V} = N \left\{ \sum_{k_s \in \mathcal{S}} \eta_{k_s}^2 \frac{\widehat{\sigma}_{k_s}^2}{n_{k_s}} + \sum_{k_t \in \mathcal{T}} \widehat{\mathbf{h}}_{k_t}(\eta)^{\top} \frac{\widehat{\Sigma}_{k_t}}{n_{k_t}} \widehat{\mathbf{h}}_{k_t}(\eta) \right\} \quad (12)$$

and the $1 - \alpha/2$ quantile for the standard normal distribution $\mathcal{Z}_{\alpha/2}$, we construct the $(1 - \alpha) \times 100\%$ confidence interval (CI)

$$\widehat{C}_{\alpha} = \left[\widehat{\Delta}_{\mathcal{T},\text{FACE}} - \sqrt{\widehat{V}/N} \mathcal{Z}_{\alpha/2}, \widehat{\Delta}_{\mathcal{T},\text{FACE}} + \sqrt{\widehat{V}/N} \mathcal{Z}_{\alpha/2} \right]. \quad (13)$$

The full FACE workflow is summarized in Algorithm 1.

Algorithm 1: FACE under generic model specifications

Data: J Target sites $k_t \in \mathcal{T}$, K Source sites $k_s \in \mathcal{S}$, and a Leading AC

- 1 **for** Target sites $k_t \in \mathcal{T}$ **do**
- 2 Estimate α_{k_t} , β_{a,k_t} to calculate the initial TATE $\widehat{\Delta}_{\mathcal{T},k_t}$, its augmentation $\widehat{\delta}_{\mathcal{T},k_t}$, and the variance estimator $\widehat{\Sigma}_{k_t}$, and transfer to the leading AC. Calculate $\widehat{\psi}_{k_t}$ and broadcast to source sites along with $\widehat{\beta}_{a,k_t}$.
- 3 **end**
- 4 **for** Source sites $k_s \in \mathcal{S}$ **do**
- 5 Estimate γ_{k_t,k_s} and α_{k_s} to calculate the site-specific augmentation $\widehat{\delta}_{\mathcal{T},k_s}$ and transfer to the leading AC. Calculate $\widehat{\sigma}_{k_s}^2$, $\widehat{\mathbf{d}}_{k_t,k_s}$ and transfer to the leading AC.
- 6 **end**
- 7 **for** Leading AC **do**
- 8 Estimate η by solving the penalized regression in (11). Construct the final global estimator as $\widehat{\Delta}_{\mathcal{T},\text{FACE}}$ by (9). Calculate the global estimator variance by (12) and construct 95% CI.
- 9 **end**

Result: Global TATE estimate, $\widehat{\Delta}_{\mathcal{T},\text{FACE}}$ and 95% CI

Remark 7. Our aggregation procedure is communication-efficient and privacy-protected, whereas aggregation procedures given in the current literature such as those in Cheng and Cai (2021) require sharing individual-level influence functions. Equation (11) is constructed using summary statistics, which provides a federated learning solution when individual-level data sharing is forbidden.

3.4. Cross-Validation and Tuning Parameters

To choose an optimal tuning parameter λ , we propose a sample splitting approach that does not require sharing individual-level data. In each site, the data is first split into training and validation datasets, keeping the same proportion within each site. In the

training datasets, [Algorithm 1](#) is implemented to obtain the summary statistics ($\tilde{\Sigma}_{k_t}$, $\tilde{\mathbf{d}}_{k_s}$, $\tilde{\sigma}_{k_s}^2$, $\tilde{\delta}_{\mathcal{T},k_s}$, and $\tilde{\delta}_{\mathcal{T},\mathcal{T}}$) needed for (11). The AC selects a grid of λ values and calculates $\hat{\eta}(\lambda)$ by solving the penalized regression in (11). In parallel, the validation datasets are used to obtain summary statistics denoted by ($\tilde{\Sigma}_{k_t}$, $\tilde{\mathbf{d}}_{k_s}$, $\tilde{\sigma}_{k_s}^2$, $\tilde{\delta}_{\mathcal{T},k_s}$ and $\tilde{\delta}_{\mathcal{T},\mathcal{T}}$). These summary statistics are calculated using the validation datasets and plugging in the parameters estimated from the corresponding training datasets. The AC sets the value of the optimal tuning parameter, λ_{opt} , to be the value corresponding to $\hat{\eta}$ that minimizes $Q(\hat{\eta})$ in the validation datasets, defined as

$$Q(\hat{\eta}) = N^V \left[\sum_{k_s \in \mathcal{S}} \hat{\eta}_{k_s}^2 \frac{\tilde{\sigma}_{k_s}^2}{n_{k_s}^V} + \sum_{k_t \in \mathcal{T}} \tilde{\mathbf{h}}_{k_t}(\hat{\eta})^\top \frac{\tilde{\Sigma}_{k_t}}{n_{k_t}^V} \tilde{\mathbf{h}}_{k_t}(\hat{\eta}) \right],$$

where N^V , $n_{k_s}^V$, and $n_{k_t}^V$ are the sample sizes for validation data from all sites, source sites $k_s \in \mathcal{S}$, and target sites $k_t \in \mathcal{T}$, respectively.

Remark 8. The upper and lower bounds on the grid of λ values can be left unrestricted; in practice, we have found that searching between 0.01 to 100 to be sufficiently large to provide good finite sample performance. For increased stability to the choice of λ , we have implemented 5-fold cross-validation, where we take λ_{opt} to be the value corresponding to $\hat{\eta}$ that minimizes the average of $Q(\cdot)$ over 5-folds (Chernozhukov et al. 2018).

3.5. FACE-X Incorporating Machine Learning

Given recent advances in incorporating machine learning methods for causal inference (Chernozhukov et al. 2018), we propose a cross-fitted FACE, termed FACE-X, which allows for the incorporation of flexible nonparametric machine learning methods in estimating nuisance functions such as the OR, PS, and density ratio models. Unlike the base FACE method that relies on parametric models, FACE-X operates under weaker assumptions, as outlined in Assumption S2 in the Supplement. Specifically, we only require that the true nuisance models and their estimators are bounded, that nonparametric estimators of these models are consistent, and that the combined estimation errors satisfy a product of rates condition. This leads to a rate double robustness property, ensuring that our estimator remains consistent and asymptotically normal (Theorem S1 in the supplement) even when some nuisance estimators converge slowly, thereby requiring weaker conditions than those needed for weak double robustness. Furthermore, both the target site only estimator and the source site augmented estimator admit asymptotic approximations (Lemma S1 in the supplement). These theoretical guarantees enhance the practical applicability of FACE-X in federated learning settings with complex data structures. The detailed estimation procedures and variance structures for aggregation are provided in supplement B.

4. Theoretical Guarantees

In this section, we provide the theoretical results for the FACE estimator. We start with a high-level theory for a generic choice of models in [Section 4.1](#). Then, we discuss the efficiency

gained from leveraging source sites in [Section 4.2](#). In our asymptotic theory, N is allowed to grow but the distribution for $(Y, \mathbf{X}^\top, A, R)^\top$ and $J + K$ are fixed.

4.1. Theory for General FACE

To compress notation, we combine the broadcast parameters and their asymptotic limits as

$$\begin{aligned} \bar{\theta}_{k_t} &= (\bar{\psi}_{k_t}^\top, \bar{\beta}_{1,k_t}^\top, \bar{\beta}_{0,k_t}^\top)^\top, \\ \bar{\theta}_{k_t} &= (\mathbb{E}\{\psi(\mathbf{X})^\top \mid R = k_t\}, \bar{\beta}_{1,k_t}^\top, \bar{\beta}_{0,k_t}^\top)^\top. \end{aligned} \quad (14)$$

Assumption 2. For absolute constants $M, \varepsilon > 0$,

(a) (Regularity of estimators) The estimators $\hat{M}_{\mathcal{T}}$, $\hat{\delta}_{\mathcal{T},k_t}$, $\hat{\beta}_{a,k_t}$, and $\hat{\delta}_{\mathcal{T},k_s}$ admit the following asymptotically linear representations

$$\begin{aligned} \sqrt{N_{\mathcal{T}}}(\hat{M}_{\mathcal{T}} - \bar{M}_{\mathcal{T},\mathcal{T}}) &= \frac{1}{\sqrt{N_{\mathcal{T}}}} \sum_{k_t \in \mathcal{T}} \sum_{i \in \mathcal{I}_{k_t}} \zeta_i + o_p(1), \\ \sqrt{N_{\mathcal{T}}}(\hat{\delta}_{\mathcal{T},\mathcal{T}} - \bar{\delta}_{\mathcal{T},\mathcal{T}}) &= \frac{1}{\sqrt{N_{\mathcal{T}}}} \sum_{k_t \in \mathcal{T}} \sum_{i \in \mathcal{I}_{k_t}} \xi_{i,\mathcal{T}} + o_p(1), \\ \sqrt{n_{k_s}}(\hat{\delta}_{\mathcal{T},k_s} - \bar{\delta}_{\mathcal{T},k_s}) &= \frac{1}{\sqrt{n_{k_s}}} \sum_{i \in \mathcal{I}_{k_s}} \xi_{i,k_s} \\ &\quad + \sqrt{n_{k_s}} \sum_{k_t \in \mathcal{T}} \bar{\mathbf{d}}_{k_t,k_s}^\top (\hat{\theta}_{k_t} - \bar{\theta}_{k_t}) + o_p(1), \\ \sqrt{n_{k_t}}(\hat{\beta}_{a,k_t} - \bar{\beta}_{a,k_t}) &= \frac{1}{\sqrt{n_{k_t}}} \sum_{i \in \mathcal{I}_{k_t}} \mathbf{v}_{i,a} + o_p(1). \end{aligned}$$

with bounded asymptotic limits $\bar{M}_{\mathcal{T},\mathcal{T}}$, $\bar{\delta}_{\mathcal{T},\mathcal{T}}$, $\bar{\delta}_{\mathcal{T},k_s}$, $\bar{\mathbf{d}}_{k_t,k_s}$ and iid mean zero random variables $\zeta_i, \xi_{i,\mathcal{T}}, \xi_{i,k_s}$.

(b) (Compact support) The covariates $\mathbf{X}$ and their functions $\psi(\mathbf{X})$ in the density ratio are in compact sets $\mathbf{X} \in [-M, M]^p$ and $\psi(\mathbf{X}) \in [-M, M]^q$ almost surely.

(c) (Stable variance) The variance of ξ_{i,k_s} is in the set $[\varepsilon, M]$. The variance-covariance matrix

$$\Sigma_{k_t} = \text{var} \left\{ \left(\zeta_i, \xi_{i,\mathcal{T}}, \psi(\mathbf{X}_i)^\top, \mathbf{v}_{i,1}^\top, \mathbf{v}_{i,0}^\top \right)^\top \mid R = k_t \right\}$$

has eigenvalues all in $[\varepsilon, M]$ for some positive constant ε and M .

(d) (Regularity of auxiliary estimators) The estimators $\hat{\Sigma}_{k_t}$, $\hat{\sigma}_{k_s}^2$, $\hat{\mathbf{d}}_{k_s}$ are $\sqrt{N}$ -consistent

$$\begin{aligned} \sum_{k_t \in \mathcal{T}} \|\hat{\Sigma}_{k_t} - \Sigma_{k_t}\| + \sum_{k_s \in \mathcal{S}} \left\{ \left| \hat{\sigma}_{k_s}^2 - \text{var}(\xi_{i,k_s} \mid R_i = k_s) \right| \right. \\ \left. + \|\hat{\mathbf{d}}_{k_s} - \bar{\mathbf{d}}_{k_s}\| \right\} = O_p(N^{-1/2}). \end{aligned}$$

(e) (Root- N rate consistency) For each target site $k_t \in \mathcal{T}$, at least one of the two models is correctly specified:

-i the PS model is consistently estimated:

$$\begin{aligned} \sup_{a=0,1} \sup_{\|\mathbf{x}\|_\infty \leq M} \sum_{k_t \in \mathcal{T}} |\mathbb{P}(A = a \mid \mathbf{X} = \mathbf{x}, R = k_t) \\ - \pi_k(a, \mathbf{x}; \hat{\alpha}_{k_t})| = O_p(N^{-1/2}). \end{aligned}$$

-ii the OR model is consistently estimated:

$$\sup_{a=0,1} \sup_{\|\mathbf{x}\|_\infty \leq M} \sum_{k_t \in \mathcal{T}} |\mathbb{E}(Y | A = a, \mathbf{X} = \mathbf{x}, R = k_t) - m_{k_t}(a, \mathbf{x}; \hat{\beta}_{a,k_t})| = O_p(N^{-1/2}).$$

Assumptions 2(a) and **2(e)** are the typical regularity conditions and can be verified in two steps: (a) asymptotic normality of model estimators (Van der Vaart 2000) and (b) local expansion of the estimators. **Assumption 2(c)** regulates the scale of variability of the data, which leads to a stable variance for $\hat{\Delta}_{\mathcal{T},\text{FACE}}$. **Assumption 2(e)** ensures identification of the true TATE by anchoring on $\hat{\Delta}_{\mathcal{T},\mathcal{T}}$ (Bang and Robins 2005). Note that in the setting of multiple target sites, **Assumption 2(e)** allows for each target site to have different correct model specifications for either the OR model or the PS model. In supplement C, we provide a detailed set of conditions corresponding to the realization of logistic regression models to estimate nuisance models.

We now state the theory for the general FACE estimation.

Theorem 1. Under **Assumptions 1** and **2**, the FACE estimator is consistent and asymptotically normal with consistent variance estimation $\hat{\mathcal{V}}$,

$$\sqrt{N/\hat{\mathcal{V}}}(\hat{\Delta}_{\mathcal{T},\text{FACE}} - \Delta_{\mathcal{T}}) \rightsquigarrow \mathcal{N}(0, 1).$$

We use $\rightsquigarrow$ for convergence in distribution.

Theorem 1 implies that (13) provides asymptotically honest coverage.

Corollary 1. Under **Assumptions 1** and **2**, the coverage rate of the confidence interval (13) approaches the nominal level asymptotically

$$\lim_{N \rightarrow \infty} \mathbb{P}(\Delta_{\mathcal{T}} \in \hat{\mathcal{C}}_\alpha) = 1 - \alpha.$$

The proof of **Theorem 1** and **Corollary 1** is given in Supplement D.3. A key step in the proof of **Theorem 1** is the analysis of the L_1 penalized adaptive selection and aggregation (11). We are able to establish the oracle property (Fan and Li 2001a), that is, the data-driven selection and aggregation through (11) is asymptotically equivalent to the process with a priori selection and optimal aggregation. The problem is different from the typical penalized regression, so we develop a new proof strategy. We first analyze the optimal combination with oracle selection, in which the biased augmentations are excluded. For unbiased augmentations, $\hat{\Delta}_{\mathcal{T},k_s} - \hat{\Delta}_{\mathcal{T},\mathcal{T}} = O_p(N^{-1/2})$, the penalty term is asymptotically negligible, that is, $\lambda(\hat{\Delta}_{\mathcal{T},k_s} - \hat{\Delta}_{\mathcal{T},\mathcal{T}})^2 = o_p(N^{-1/2})$ when λ is chosen such that $\lambda \asymp N^\nu$ with $\nu \in (0, 1/2)$. Thus, the estimated combination converges to the asymptotic limit at the regular $N^{-1/2}$ rate. Finally, we show that the estimated combination with oracle selection also solves the original problem with high probability.

Remark 9. The proposed FACE estimator has CI with honest coverage of the true TATE if all the biases are detectable $|\bar{\delta}_{\mathcal{T},k_s} - \bar{\delta}_{\mathcal{T},\mathcal{T}}| \gg N^{-1/2}$. However, in accordance with the limit characterized by the information lower bound, it is not possible

to detect source sites with small biases of $|\bar{\delta}_{\mathcal{T},k_s} - \bar{\delta}_{\mathcal{T},\mathcal{T}}| \lesssim N^{-1/2}$ order. Involving these sites in the final TATE will introduce non-negligible bias that cannot be corrected. With the presence of weakly biased $\hat{\delta}_{\mathcal{T},k_s}$, the undetectable bias may induce a biased, nonregular FACE estimator, as well as undercoverage of the confidence interval. While such an issue would not occur in the large N and finite K framework of our theory, we offer a possible remedy for regularity with regard to finite sample performance. In the aggregation (11), we may substitute the penalty factor with a truncated Wald statistic

$$\hat{\eta} = \arg \min_{\eta \in \mathbb{R}^K} \hat{L}(\eta) + \lambda \sum_{k_s \in \mathcal{S}} |\eta_{k_s}| \left\{ \left(\frac{\sqrt{N} |\hat{\delta}_{\mathcal{T},k_s} - \hat{\delta}_{\mathcal{T},\mathcal{T}}|}{\hat{\sigma}_{\delta,k_s}} - 1/\lambda \right) \vee 0 \right\}, \quad N^{-1/2} \ll \lambda \ll 1,$$

where $\hat{\sigma}_{\delta,k_s}^2$ is the variance estimator for the $\hat{\delta}_{\mathcal{T},k_s} - \hat{\delta}_{\mathcal{T},\mathcal{T}}$,

$$\hat{\sigma}_{\delta,k_s}^2 = N \left(\eta_{k_s}^2 \frac{\hat{\sigma}_{k_s}^2}{n_{k_s}} + \sum_{k_t \in \mathcal{T}} \hat{\mathbf{h}}_{k_t,k_s}^\top \frac{\hat{\Sigma}_{k_t}}{n_{k_t}} \hat{\mathbf{h}}_{k_t,k_s} \right),$$

$$\hat{\mathbf{h}}_{k_t,k_s} = \left(0, -\frac{n_{k_t}}{N_{\mathcal{T}}}, \hat{\mathbf{d}}_{k_t,k_s}^\top \right)^\top.$$

Using the modified penalty factor that converges to its limit at $N^{-1/2}$ rate for $k_s \in \mathcal{S}^\dagger = \{k_s : |\bar{\delta}_{\mathcal{T},k_s} - \bar{\delta}_{\mathcal{T},\mathcal{T}}| \lesssim \lambda^{-1} N^{-1/2}\}$ and diverges to $+\infty$ elsewhere, we will be able to establish $N^{-1/2}$ concentration of $\hat{\eta}$ at

$$\tilde{\eta} = \arg \min_{\eta \in \mathcal{S}^\dagger} L^*(\eta) + \lambda \sum_{k_s \in \mathcal{S}} |\eta_{k_s}| \left\{ \left(\frac{\sqrt{N} |\bar{\delta}_{\mathcal{T},k_s} - \bar{\delta}_{\mathcal{T},\mathcal{T}}|}{\bar{\sigma}_{\delta,k_s}} - 1/\lambda \right) \vee 0 \right\}$$

over the support $\mathcal{S}^\dagger$. The approximation in Lemma S6 will hold, as the extra bias term decays to zero

$$\sum_{k_s \in \mathcal{S}^\dagger} |(\hat{\eta}_{k_s} - \tilde{\eta}_{k_s}) \sqrt{N} (\bar{\delta}_{\mathcal{T},k_s} - \bar{\delta}_{\mathcal{T},\mathcal{T}})| = O_p(\bar{\delta}_{\mathcal{T},k_s} - \bar{\delta}_{\mathcal{T},\mathcal{T}}) = o_p(1).$$

Consequently, the modified FACE estimator would be asymptotically normal after removing the bias. The confidence interval would have reasonable coverage if the accumulated bias $\sum_{k_s \in \mathcal{S}^\dagger} \tilde{\eta}_{k_s} (\bar{\delta}_{\mathcal{T},k_s} - \bar{\delta}_{\mathcal{T},\mathcal{T}})$ is smaller than the standard deviation.

Remark 10. We may modify the CI to conservatively capture the uncertainty of $\hat{\Delta}_{\mathcal{T},\text{FACE}}$. Denote the estimators needed for aggregation as $\hat{\theta}$, including $\hat{M}_{\mathcal{T}}, \hat{\delta}_{\mathcal{T},\mathcal{T}}, \hat{\delta}_{\mathcal{T},k_s}, \hat{\Sigma}_{k_t}, \hat{\mathbf{h}}_{k_t}$. We denote the process of getting $\hat{\Delta}_{\mathcal{T},\text{FACE}}$ from $\hat{\theta}$ as $\hat{\Delta}_{\mathcal{T},\text{FACE}} = \mathcal{H}(\hat{\theta})$, which is continuous along $\hat{\theta}$ and it is deterministic given $\hat{\theta}$. Suppose θ_* is the asymptotic limit of $\hat{\theta}$. Based on the standard asymptotic normality of $\sqrt{N}(\hat{\theta} - \theta_*)$, we may construct a standard confidence region $\mathcal{C}(\hat{\theta}, \alpha)$ such that

$$\liminf_{N \rightarrow \infty} \mathbb{P}\{\theta_* \in \mathcal{C}(\hat{\theta}, \alpha)\} \geq 1 - \alpha.$$

Mapping through $\mathcal{H}$ will produce a confidence interval for $\mathcal{A}(\theta_*)$,

$$\liminf_{N \rightarrow \infty} \mathbb{P}\{\mathcal{H}(\theta_*) \in \mathcal{H} \circ \mathcal{C}(\hat{\theta}, \alpha)\} \geq 1 - \alpha.$$

This process will account for the uncertainty in site selection but may produce a conservative CI.

Remark 11. For consistency of $\hat{\Delta}_{\mathcal{T},\text{FACE}}$, we require that the PS or OR model is correct for the target sites but allow the models for the source sites and density ratio to be misspecified. To meaningfully leverage information from source sites for the TATE, we would expect that many $k_s \in \mathcal{S}$ among the source sites (i) satisfy the ignorability [Condition 1\(d\)](#) and (ii) either the OR model $m(a)$ is correct, or both the PS π_{k_s} and the density ratio ω_{k_t, k_s} models are correct. For source sites satisfying the conditions above, their site-specific augmentations are unbiased and thus contribute to the efficiency improvement of $\hat{\Delta}_{\mathcal{T},\text{FACE}}$.

Remark 12. When the oracle property can be achieved for estimating sparse $\hat{\beta}_{a,k}$, $\hat{\gamma}_{k_t, k_s}$, and $\hat{\alpha}_k$ under the high-dimensional setting (p grows with n_k) with minimal signal strength (Fan and Li 2001b; Zou 2006; Lv and Fan 2009), it ensures the exact identification of the nonzero elements in the coefficients, which reduces the high-dimensional setting to the low-dimensional setting studied in our paper. For the general high-dimensional setting without guarantee of the oracle property, our FACE method can be extended under ideal situations with perfectly specified models through cross-fitting (Chernozhukov et al. 2018). Extensions beyond such an ideal setting require careful investigation beyond the approaches considered in the existing literature since general Neyman orthogonality requires all estimated models to be consistent for their target conditional expectations. As a fundamental issue in learning the TATE, the potentially incompatible source sites (i.e., source sites in $\mathcal{S} \setminus \mathcal{S}^*$) will invalidate such a consistency requirement.

4.2. Relative Efficiency

Notice that we recover the initial TATE estimator $\hat{\Delta}_{\mathcal{T},\mathcal{T}}$ from (9) if $\hat{\eta} = \mathbf{0}$. Since we are minimizing the post-aggregation variance, the optimal solution must be no worse than any alternative solutions. If there exists informative source sites in $\mathcal{S}'$, as defined in [Assumption 3](#), improvement in the efficiency of FACE compared to the target only estimator is guaranteed.

Assumption 3. For a nonempty set $\mathcal{S}' \subseteq \mathcal{S}$,

(a) One of the following holds:

(i) Correct OR: the OR model is consistently estimated:

$$\sup_{a=0,1} \sup_{\|\mathbf{x}\|_\infty \leq M} \sum_{k_t \in \mathcal{T}} |\mathbb{E}(Y | A = a, \mathbf{X} = \mathbf{x}, R = k_t) - m_{k_t}(a, \mathbf{x}; \hat{\beta}_{a, k_t})| = O_p(N^{-1/2});$$

(ii) Consistent weighting: the PS and density ratio models are consistently estimated:

$$\begin{aligned} & \sup_{a=0,1} \sup_{\|\mathbf{x}\|_\infty \leq M} \sum_{k_s \in \mathcal{S}'} |\mathbb{P}(A = a | \mathbf{X} = \mathbf{x}, R = k_s) - \pi_{k_s}(a, \mathbf{x}; \hat{\alpha}_{k_s})| \\ & + \sum_{k_t \in \mathcal{T}} \sum_{k_s \in \mathcal{S}'} \left| \frac{\mathbb{P}(R = k_t | \mathbf{X} = \mathbf{x}) \mathbb{P}(R = k_s)}{\mathbb{P}(R = k_s | \mathbf{X} = \mathbf{x}) \mathbb{P}(R = k_t)} \right. \\ & \left. - \omega_{k_t, k_s}(\mathbf{x}; \hat{\gamma}_{k_t, k_s}) \right| = O_p(N^{-1/2}). \end{aligned}$$

(b) Informative source: Let $\boldsymbol{\vartheta} = (\boldsymbol{\psi}(\mathbf{X})^\top, \mathbf{v}_1^\top, \mathbf{v}_0^\top)^\top$ be the combined influence function for broadcast estimators. For all $k_s \in \mathcal{S}'$

$$\left| \text{cov} \left(\frac{\zeta + \xi_{\mathcal{T}}}{\mathbb{P}(R \in \mathcal{T})}, -\frac{\xi_{\mathcal{T}}}{\mathbb{P}(R \in \mathcal{T})} + \sum_{k_t \in \mathcal{T}} \frac{\mathbb{I}(R = k_t)}{\mathbb{P}(R = k_t)} \boldsymbol{\vartheta}^\top \bar{\mathbf{d}}_{k_t, k_s} | R \in \mathcal{T} \right) \right| \geq \varepsilon.$$

The two model consistency conditions in [Assumption 3\(a\)](#) ensure the consistency of the doubly robust estimator $\hat{\Delta}_{\mathcal{T}, k_s}$. [Assumption 3\(b\)](#) characterizes the informativeness of a source site k_s such that the updated direction $(\hat{\delta}_{\mathcal{T}, k_s} - \hat{\delta}_{\mathcal{T}, \mathcal{T}})$ is correlated with the initial $\hat{\Delta}_{\mathcal{T}, \mathcal{T}}$. The covariance in the condition is likely to be negative with the opposite sign of $\xi_{\mathcal{T}}$.

Proposition 1. Under the conditions of [Theorem 1](#), the asymptotic variance of $\hat{\Delta}_{\mathcal{T}, \text{FACE}}$ is no larger than that of $\hat{\Delta}_{\mathcal{T}, \mathcal{T}}$. Moreover, if [Assumption 3](#) holds, the asymptotic variance of $\hat{\Delta}_{\mathcal{T}, \text{FACE}}$ is strictly smaller than that of $\hat{\Delta}_{\mathcal{T}, \mathcal{T}}$.

The proof is given in supplement D.5. [Proposition 1](#) offers a guarantee on the relative efficiency in general settings. As the exact efficiency gain may take different forms under general settings, we showcase the efficiency gain with a clear interpretation under a simple ideal setting in supplement D.7.

Remark 13. While the federated estimator in FACE leverages data from source sites to enhance efficiency, the extent of this improvement is inherently constrained by the necessity to estimate appropriate aggregation weights. Consequently, even if the size of the source data grows infinitely large, the uncertainty in the estimator cannot diminish beyond the limit imposed by the finite size of the target site. This limitation arises because the estimation of weights introduces additional variability that does not vanish with increasing source data size. In contrast, when inverse variance weights (IVW) are known—such as in scenarios where the source and target populations are identical—the efficiency improvement can indeed grow indefinitely as the source data size increases. Therefore, there exists a tradeoff between robustness and efficiency in FACE: using estimated weights enhances robustness to distributional differences between source and target data but limits potential efficiency gains compared to using known IVW under identical distributions, although the latter is a strong, typically untenable assumption.

5. Simulation Studies

We study the finite sample performance of (i) the FACE estimator against four estimators: (ii) an estimator that leverages target data only (target-only), (iii) a sample-size adjusted estimator (SS), (iv) an inverse-variance weighted estimator (IVW), and (v) an exponentially-tilted augmented inverse probability weighted estimator (Tilted-AIPW) that multiplies density-ratio weighted site-specific AIPW estimators and aggregates via SS. We examine the bias, root mean

Table 1. Eight levels of deviation corresponding to the informativeness of source sites where the true TATE is $\Delta_\tau = 3.0$.

Deviation	Description of source sites	Source ATEs $\Delta_k, k = 2, \dots, 10$
1	All source sites informative	$\Delta_2, \dots, \Delta_{10} = 3.0$.
2	One weakly non-informative source site	$\Delta_2 = 3.2, \Delta_3, \dots, \Delta_{10} = 3.0$.
3	Two weakly non-informative source sites	$\Delta_2 = \Delta_3 = 3.4, \Delta_4, \dots, \Delta_{10} = 3.0$.
4	Three moderately non-informative source sites	$\Delta_2 = \Delta_3 = \Delta_4 = 3.6, \Delta_5, \dots, \Delta_{10} = 3.0$.
5	Five moderately non-informative source sites	$\Delta_2, \dots, \Delta_6 = 3.8, \Delta_7, \dots, \Delta_{10} = 3.0$.
6	Seven moderately non-informative source sites	$\Delta_2, \dots, \Delta_8 = 4.0, \Delta_9 = \Delta_{10} = 3.0$.
7	Eight strongly non-informative source sites	$\Delta_2, \dots, \Delta_9 = 4.5, \Delta_{10} = 3.0$.
8	All source sites strongly non-informative	$\Delta_2, \dots, \Delta_{10} = 5.0$.

square error (RMSE), coverage probability of the 95% CIs, and length of the 95% CIs of these estimators across 500 simulations.

5.1. Data Generation

We set $J + K = 10$ sites, with the first site as the target and $K = 9$ source sites. We set the sample size $n_k = 200, k = 1, \dots, 10$. In supplement E, we include results for $n_k = 400, k = 1, \dots, 10$ and vary the number of sites $J + K \in \{10, 20, 40\}$. To explore the effect of model misspecification, we vary the misspecification of the true OR, PS, and density ratio models. To allow for heterogeneity in the covariate distribution between sites, the covariates in each site $\mathbf{X}_{kp}$ are generated from a skewed normal distribution, $\mathbf{X}_{kp} \sim \mathcal{SN}(\mathbf{x}; \kappa_{kp}, \phi_{kp}^2, \nu_{kp})$, where $k = 1, \dots, J + K$ indexes the sites and $p = 1, \dots, 10$ indexes the 10 covariates, κ_{kp} is the location parameter, ϕ_{kp} is the scale parameter, and ν_{kp} is the skewness parameter. For all sites, we let $\kappa_k \in (0.10, 0.15)$ and $\phi_k = (1, \dots, 1)$. For the target site, we set $\nu_k = 0$. For the source sites, we let $\nu_k \in [0, 0.2]$ so that the exponential tilt model provides varying approximation quality for projecting the covariate distributions to the target site.

The true potential outcomes are generated as

$$Y_k(a) = [(\mathbf{X}_k - \mu_1)^\top, (\mathbf{X}_k^{\circ 2})^\top] (\boldsymbol{\beta}_{1a}^\top, \boldsymbol{\beta}_{2a}^\top)^\top + \Delta_k I(a = 1) + \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(0, 2\sqrt{5}), \quad a = 0, 1,$$

where $\mathbf{X}_k^{\circ 2}$ denotes $\mathbf{X}_k$ squared element-wise, $\boldsymbol{\beta}_{11} = (0.4, \dots, 1.2)$, and $\boldsymbol{\beta}_{10} = (0.4, \dots, 1.2)$ with equally-spaced increments for a length 10, and Δ_k is the ATE. We consider eight different settings where the level of deviation varies, ranging from all source sites being informative to all being strongly non-informative. Table 1 describes in each setting how similar $\Delta_k, k \in \mathcal{S}$ are to the target $\Delta_\tau = 3.0$.

The true PS model is generated as $A_k | \mathbf{X} = \mathbf{x} \sim \text{Bernoulli}(\pi_k), \pi_k = \text{expit}(\mathbf{X}_k \boldsymbol{\alpha}_{1k} + \mathbf{X}_k^{\circ 2} \boldsymbol{\alpha}_{2k})$, where for the target and source sites, $\boldsymbol{\alpha}_{1k} = (0.5, \dots, -0.5)$, with equally-spaced decrements for a length 10 and $\boldsymbol{\alpha}_{2k} = (-0.5, 0, \dots, 0)$. For all sites, we fit linear regression models for the OR and logistic regression models for the PS, where we misspecify models by only including the linear terms of the covariates $\mathbf{X}_k$.

5.2. Simulation Settings

We consider the following settings. In Setting 1, we examine the scenario where the OR and PS models are correctly specified, but the density ratio models are misspecified. In Setting 2, we misspecify the OR while keeping the PS and density ratio models

correctly specified. In Setting 3, we misspecify the PS but correctly specify the OR model and density ratio models. In each setting, we choose the tuning parameter λ by the distributed cross-validation procedure described in Section 3.4, where we split the simulated datasets in each site into two equally sized training and validation datasets and take the optimal λ over 5-folds.

5.3. Simulation Results

In Figure 1, we summarize the bias, RMSE, coverage, and length of the 95% CIs of FACE and the four alternative estimators in Setting 1 where only the density ratio models are misspecified. Results for Settings 2 and 3 with model misspecification are provided in supplement E. When all source sites are informative (deviation level 1), all estimators perform well. When only one (level 2) or two (level 3) source sites are weakly non-informative, FACE, SS, and IVW perform well, with minimal bias, RMSE smaller than the target-only estimator, nominal coverage, and substantially shorter average CIs compared to the target-only. However, as the proportion and bias of non-informative source sites increase, only FACE shows relatively good robustness against negative transfer with minimal bias, RMSE below that of target-only, nominal coverage, and shorter average CIs compared to target-only. On the other hand, we observe large biases and poor coverage for the alternative estimators. Overall, the RMSE of FACE is lower than that of the target-only estimator and approaches the target-only as the proportion and bias of non-informative source sites increase. The coverage of FACE is close to the nominal 95% across different levels of deviation and the length of the 95% CI of FACE is shorter than that of the target-only estimator when there are informative source sites.

In Figure 2, the adaptivity of the FACE weights $\hat{\eta}$ can be seen across the eight deviation levels. As the proportion of informative source sites decreases (deviation level increases), the weight given to the target site increases. This adaptivity allows for reduced bias relative to other methods such as IVW or SS weighting, which have fixed weights that contribute to the large bias and low coverage when there are non-informative source sites.

6. Comparative Effectiveness of COVID-19 Vaccines

To illustrate FACE, we study the comparative effectiveness of BNT162b2 (Pfizer) versus mRNA-1273 (Moderna) for the prevention of COVID-19 outcomes in five VA sites. It is of interest

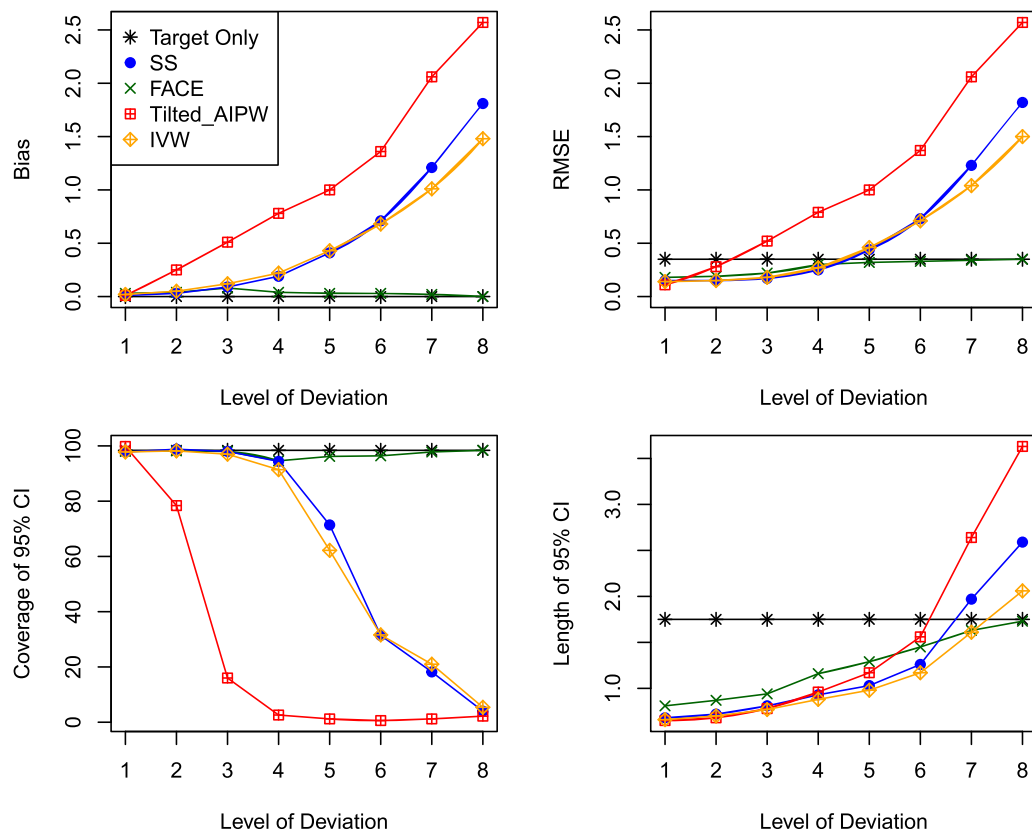


Figure 1. Setting 1. Bias, RMSE, coverage, and length of 95% CI's of the target-only, SS, FACE, exponentially tilted AIPW, and IVW estimators of the TATE across 500 simulations with misspecified density ratio models.

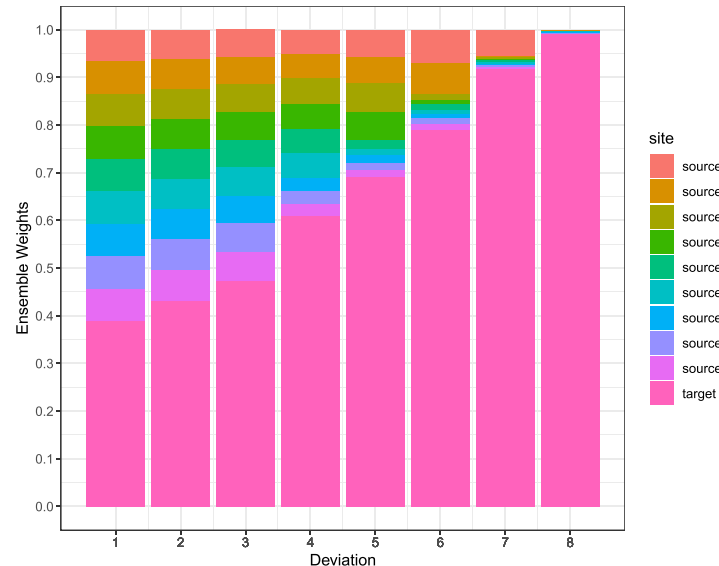


Figure 2. FACE ensemble weights $\hat{\eta}$ for each site across eight deviation levels.

to understand the real-world effectiveness of these vaccines, but head-to-head comparisons have been rare. A recent emulated target trial using the EHRs of U.S. veterans showed that the 24-week risk of COVID-19 outcomes was low for patients who received either vaccine, but lower for veterans assigned to Moderna compared to Pfizer and potentially heterogeneous across patient populations (Dickerman et al. 2021), suggesting that only reporting an overall ATE may be misleading for certain target populations. Using FACE, we examine the TATE in a federated

data setting where the target population of interest is one of five sites (North Atlantic, Southwest, Midwest, Continental, or Pacific) in the VA healthcare system. Our problem is more challenging than that of Dickerman et al. (2021) or Lin et al. (2022) due to the federated data setting and the different target populations of interest that we are able to study.

Inclusion criteria included veteran status, at least 18 years of age by January 1, 2021, no previously documented COVID-19 infection, no previous COVID-19 vaccination, and documented

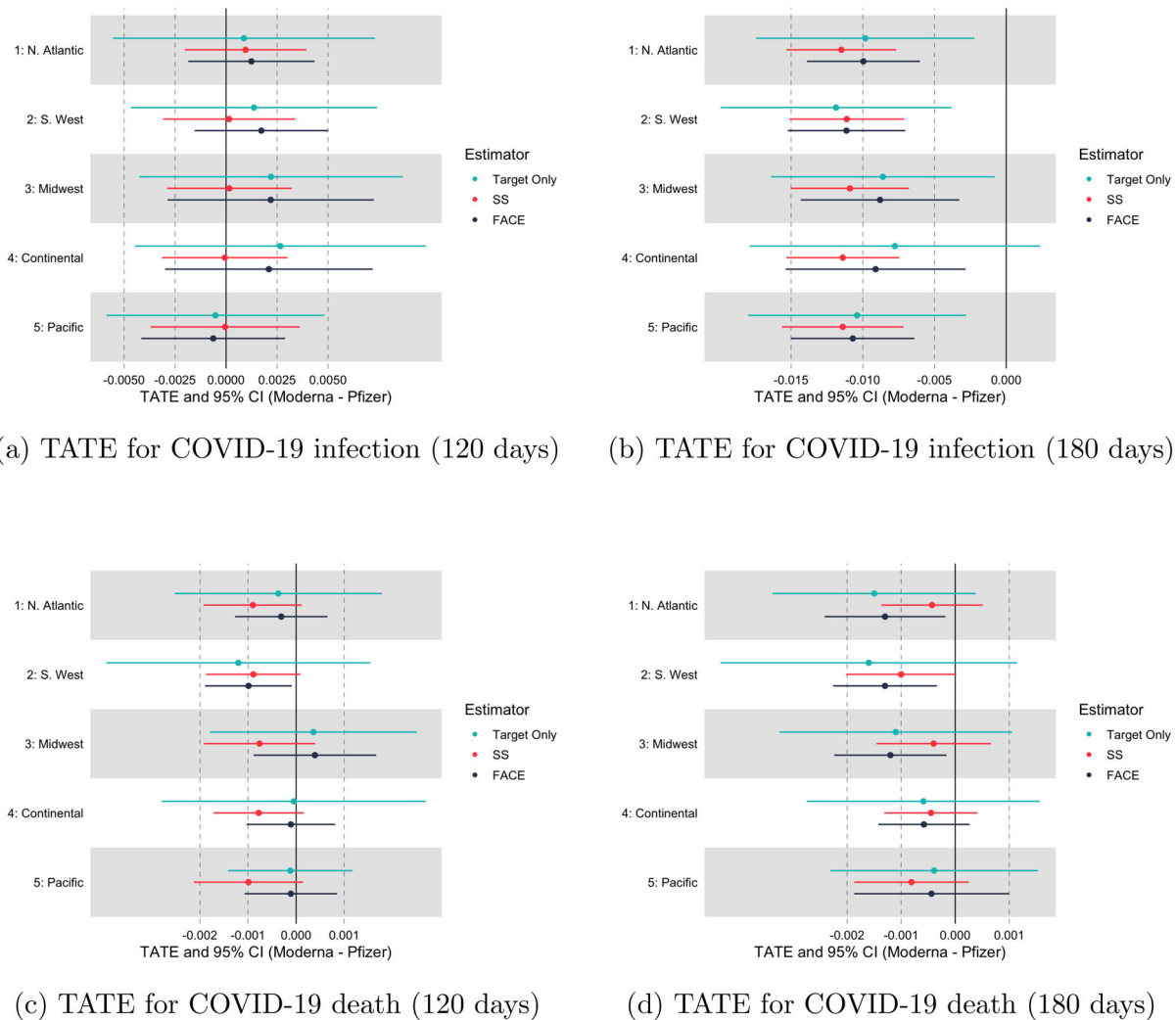


Figure 3. TATE estimates for the comparative effectiveness of Moderna versus Pfizer vaccines for four outcomes.

two-dose COVID-19 vaccination with either Pfizer or Moderna between January 1 and March 24, 2021. For each eligible veteran, follow-up began on the day that the second dose of vaccine was received (baseline) and ended on the day of death, 120 or 180 days after baseline, or the end of the study time period (September 24, 2021). The outcomes of interest were documented SARS-CoV-2 infection either 120 or 180 days after baseline and death with COVID-19 infection either 120 or 180 days after baseline.

Among the 608,359 eligible veterans, 293,137 (48.2%) received Pfizer and 315,222 (51.8%) received Moderna. Baseline characteristics among the two groups were similar within sites. Across sites, there was heterogeneity in race (a larger proportion of Asians in the Pacific), and ethnicity (a larger Hispanic population in the Southwest and Pacific). Baseline characteristics in each of the five sites are summarized in supplementary Tables S3 and S4. All models were adjusted for age, sex, race, ethnicity, residence, and important comorbidities: chronic lung disease (including asthma, bronchitis, and chronic obstructive pulmonary disease), cardiovascular disease (including acute myocardial infarction, cardiomyopathy, coronary heart disease, heart failure, and peripheral vascular disease), hypertension, type 2 diabetes, chronic kidney disease,

autoimmune diseases (including HIV infection, rheumatoid arthritis, etc.), and obesity (defined as body mass index of 30 or greater).

The raw event rates for documented COVID-19 infection within 180 days of receiving the second dose for Pfizer (Moderna) in the five sites were 2.81% (1.93%) in the North Atlantic, 3.58% (3.23%) in the Southwest, 2.25% (2.08%) in the Midwest, 2.97% (2.36%) in the Continental, and 2.80% (1.43%) in the Pacific. The raw event rates for death with COVID-19 infection within 180 days of receiving the second dose for Pfizer (Moderna) were 0.37% (0.06%) in the North Atlantic, 0.36% (0.23%) in the Southwest, 0.18% (0.21%) in the Midwest, 0.21% (0.26%) in the Continental, and 0.11% (0.09%) in the Pacific.

Figure 3 shows the TATE estimates for the four outcomes of interest: (a) 120-day COVID-19 infection, (b) 180-day COVID-19 infection, (c) 120-day death with COVID-19 infection, and (d) 180-day death with COVID-19 infection. For each outcome, the target population is taken to be one of the five sites. Three estimators are compared along with their 95% CIs: (i) target-only, (ii) a sample-size weighted estimator that leverages each site where η_k is taken to be n_k/N (SS), $k = 1, \dots, 5$, and (iii) the FACE estimator. Our results indicate that the FACE estimator

tracks the target-only estimator more closely compared to the SS estimator. Compared to the target-only estimator, the FACE estimator has substantially tighter confidence intervals, resulting in qualitatively different conclusions in certain cases, for example, 180-day COVID-19 infection in the Continental site, 120-day death with COVID-19 infection in the Southwest site, and 180-day death with COVID-19 infection in the Midwest, North Atlantic, and Southwest sites. Using FACE, our results show that veterans who received Moderna had an approximately 1% lower rate of 180-day COVID-19 infection compared to Pfizer, and this difference appeared consistent across sites.

7. Discussion

FACE can easily be generalized to the setting where some sites have RCT data, and one could define the target population as the set of trial participants. When the RCT data is treated as the anchoring site, the target site PS model is known, so the target site estimator for the TATE is consistent, and the global adaptive estimator is likely to be more reliable. FACE can thus be viewed as contributing to recent work on reducing treatment effect variance in experimental studies using observational studies (Athey, Chetty, and Imbens 2020).

FACE can also be adapted to target different causal parameters of interest, such as the average treatment effect of the treated or the conditional average treatment effect. It may also be of interest to extend FACE to model-based treatment effect metrics defined by, for example, marginal structural models or structural nested models, although further elaboration on the causal interpretations would be needed. The major advantage of using the risk difference (e.g., ATE) as a metric is its model-free property. In contrast, relative risk metrics cannot characterize the population shift in a justifiable manner. Either the causal parameter is defined conditionally on X , thus, totally independent of population shift (Vansteelandt and Joffe 2014; Hou, Bradic, and Xu 2023), or the causal parameter is defined marginally based on a single population, but such a model would no longer hold in another shifted population (Hernán, Brumback, and Robins 2001).

An interesting extension is when no outcome or treatment information is observed for the target site but predictive surrogates may exist (Han, Wang, and Cai 2022). This may be the case when it is expensive, time-consuming, or otherwise challenging to collect information on target samples (Leek et al. 2010; Ling et al. 2022). In such scenarios, it may be desirable to use data from source sites that include covariates, treatment, and outcome information. When there is substantial heterogeneity and outliers may exist, a strategy of integrative analysis is to identify a prevailing model, defined as the model satisfied by the majority of the sites. Identifying the prevailing model can be achieved via, for example, the majority rule (Hastie and Kameda 2005). Guo et al. (2023) developed a theoretically justified robust inference for federated meta-learning (RIFL) framework to construct uniformly valid confidence intervals for the unknown prevailing model using multi-source data. Future work may seek to extend RIFL to more flexible target distributions.

Supplementary Materials

The supplementary materials are divided into six sections. In Section A, we illustrate the workflow of FACE to construct a global estimator in a federated data setting. Section B describes the estimation, aggregation procedure, and theoretical guarantees for FACE- X , a cross-fitted version of FACE that allows for nonparametric estimation of nuisance functions. In Section C, we provide a mild set of sufficient conditions for the necessary regularity conditions to hold in the special case with logistic regression models for the nuisance functions and illustrate FACE under logistic regression models. In Section D, we provide proofs for the theoretical results in Section 4 of the main paper. We also showcase the efficiency gain of FACE relative to the initial TATE estimator with an exact calculation under a simple ideal setting. In Section E, we provide additional simulation results. In Section F, we provide supplementary results corresponding to the real data analysis.

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No potential conflict of interest was reported by the author(s).

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